

Exploring Multimodal Fusion for Enhanced Emotion Recognition

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Abstract—This project explores multimodal emotion recognition using the IEMOCAP dataset, exploring the interplay between multimodal features, as well as different model architectures, to enhance accuracy and robustness. Employing the TIM-Net (Temporal-aware bi-directional Multi-scale Network) as the baseline model, additional exploration entails diverse model architectures to discern their effectiveness across various modalities. Subsequently, a novel multimodal model is devised, streamlining individual modality processing through optimised pathways prior to implementing cross-modal attention mechanisms. The multimodal model demonstrates promising performance on the IEMOCAP dataset, achieving an F1 score of 0.729 and accuracy of 0.730, underscoring its potential in multimodal emotion recognition tasks.

I. INTRODUCTION

As our world gets more deeply entwined with AI systems, understanding and replicating human emotions in machines becomes not just a possibility but a necessity for creating more empathetic technological solutions. Among the key advancements in this domain, Emotion Recognition stands out as a pivotal development.

Emotion Recognition enables AI systems to identify, interpret, and respond to human emotions, enhancing the responsiveness, adaptability, and emotional intelligence of AI systems. This capability holds immense potential across diverse applications, from virtual mental health assistants capable of providing empathetic support to sophisticated customer service analytics tools [1].

Emotion recognition is vital for enhancing human-computer interaction, enabling systems to understand and respond appropriately to users' emotional states. Recent advancements in deep

learning and the availability of multimodal datasets have paved the way for more accurate and context-aware emotion recognition systems. Furthermore, the growing availability of multimodal datasets allows us to capture rich emotional cues from multiple modalities, leading to more comprehensive and accurate emotion recognition models.

II. OBJECTIVES AND SCOPE

This project aims to leverage these developments to explore multimodal emotion recognition using the IEMOCAP dataset, integrating multimodal features to enhance accuracy and robustness. It will encompass the following key activities:

1. Implementing a baseline multimodal emotion recognition model using the TIM-Net architecture [2].
2. Investigating the integration of textual and visual features to enhance model performance.
3. Experimenting with alternative deep learning architectures, including attention-based models and graph neural networks, to evaluate their efficacy relative to the established TIM-Net model.
4. Assessing the performance of the proposed models on the IEMOCAP dataset.

The next section will detail the methodology employed to achieve the outlined objectives, including data preprocessing, model implementation, and evaluation metrics used to gauge the performance of the emotion recognition models.

III. METHODOLOGY

A. Dataset

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) dataset [3], renowned for its extensive collection of naturalistic audio-visual dialogues with detailed emotional annotations, will serve as the primary dataset for both training and testing. It encompasses approximately 12 hours of recordings featuring 5 American-English speaking pairs engaged in naturalistic conversations.

To extend the research, a multimodal dataset preprocessed by Poria et al. [4] was incorporated, notable for including audio, video, and textual. This multimodal dataset comprises 120 samples, with 110 timesteps per sample, and features 100-dimensional feature vectors for textual and audio data, along with 512-dimensional vectors for visual data, totaling 712 feature vectors. Textual features were extracted using convolutional neural networks (CNNs), while audio and visual features were derived using openSMILE and 3D-CNN respectively.

However, the dataset also posed several challenges. Firstly, the risk of overfitting emerged due to the relatively small number of samples in conjunction with the increased feature dimensionality. This scenario raises concerns that the model might learn noise specific to the training data, potentially hindering its ability to generalise to new, unseen data. Additionally, a lack of visibility into the data processing stages emerged as a notable challenge. This limitation restricted the flexibility to explore alternative preprocessing methods, such as utilising embeddings from pre-trained language models for the text modality. Moreover, dataset imbalance presents another significant challenge. The multimodal dataset exhibits an imbalanced distribution, with a predominance of angry classes compared to other emotional states. Such imbalances can bias the model towards overrepresented classes, thereby compromising its predictive accuracy for less common emotional states.

Despite these constraints, this study prioritises examining interactions between different modalities over the intricacies of feature extraction.

B. Preprocessing techniques

To streamline the dataset and maintain consistency with TIMNet, certain adjustments were made, consolidating similar emotional states. Specifically, the 'happy' and 'excited' labels were merged, as were 'angry' and 'frustrated' ones. However, this also resulted in a more imbalanced dataset, with the 'Angry' class comprising 41.3% of the dataset.

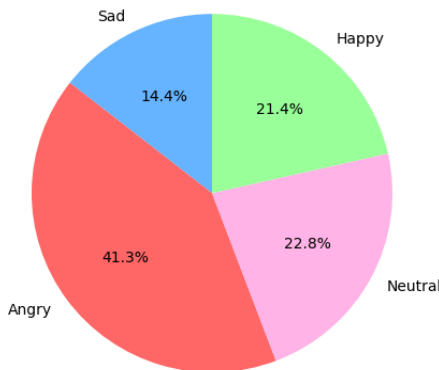


Fig. 1. Training dataset class distribution after merging 'Happy' and 'Excited', and 'Angry' and 'Frustrated' emotion classes.

To prepare the data for sequence-based, multi-class classification, several preprocessing steps were undertaken to address the issue of variable sequence lengths. Initially, label indices were shifted to allocate '0' specifically for padding. Following this, padding was applied uniformly across sequences to standardise their lengths. To differentiate between actual data and padding tokens during model training, masking was then applied, preventing the model from processing padded values and influencing predictions erroneously.

Subsequently, data batching was employed to facilitate efficient training, allowing the mask to be seamlessly integrated into each batch. Concurrently, the dataset was partitioned into training and validation sets using an 80/20 split, ensuring an adequate amount of data for both training and assessment.

Finally, the mask was applied to both the predicted values and the ground truth labels, filtering out the padded timesteps and ensuring that only the actual data points are considered for evaluation. This way, the evaluation metrics reflect the model's performance based on the relevant parts of the sequences, excluding any padded element.

C. Model Training

The TIM-Net (Temporal-aware bi-directional Multi-scale Network) [2] introduced in 2023 serves as the baseline code for this project. Alternative architectures, such as a Recurrent Neural Network (RNN), a Convolutional Neural Network (CNN), and a Transformer-based model, are explored and their performance compared with the baseline TIM-Net model.

During training, each model underwent 10 epochs of iterative learning, with each epoch comprising a complete pass through the entire dataset. A batch size of 32 was employed, balancing computational efficiency with model stability and convergence.

For optimization, the categorical cross-entropy loss function was utilised, a standard choice for multi-class classification tasks. The Adam optimizer, renowned for its efficiency and effectiveness in training deep neural networks, was employed to update model parameters iteratively.

D. Evaluation Metric

To gauge model performance, two primary metrics were employed: accuracy and F1 score. Accuracy provides a straightforward measure of the model's overall correctness, while the F1 score offers insight into its balance between precision and recall, particularly important for imbalanced datasets.

IV. EARLY EXPLORATION

Before going into model experimentation, early exploration was conducted to understand the comparative effectiveness of early versus late

fusion approaches, as well as the performance of different modalities, aiming to shed light on the optimal fusion approach and the individual contributions of audio, video, and text modalities.

For the purposes of this exploratory analysis, a straightforward architecture utilising a Bidirectional Long Short-Term Memory (BiLSTM) model was employed. The BiLSTM architecture was chosen due to its effectiveness in capturing sequential dependencies and patterns in multimodal data. This model architecture consisted of a BiLSTM layer followed by a dense layer for classification.

A. Comparison of Early and Late Fusion

Firstly, we investigated the impact of early versus late fusion architectures in handling multiple modalities.

In the early fusion approach, all modalities (audio, video, and text) were concatenated at the input level and fed directly into the BiLSTM layer. This allowed the model to jointly learn representations from all modalities simultaneously, leveraging the complementary information present in each modality from the outset. Conversely, the late fusion approach involved processing each modality individually through separate BiLSTM layers. Subsequently, the outputs from each modality were concatenated before being fed into a final BiLSTM layer for fusion. This sequential processing of modalities allowed the model to learn representations independently for each modality before combining them for classification.

Upon observation of the experimental results, it was noted that there was no discernable advantage of either fusion approach, at least within the context of the model and dataset. The accuracy and F1 score achieved by each fusion approach are summarised in Table I.

TABLE I. COMPARISON OF FUSION APPROACHES

Method	Acc.	F1
Early fusion	0.6981	0.6888
Late fusion	0.6950	0.6853

Fig. 2. Accuracy and F1 scores for each fusion approach

Despite minor differences in performance metrics between the two fusion approaches, these variances were not statistically significant. This finding suggests that, for our model and dataset, the choice between early and late fusion did not yield a substantial impact on overall performance. It's worth noting that the effectiveness of fusion strategies may vary depending on factors such as dataset characteristics, task complexity, and the inherent relationships between modalities. In light of these findings, our investigation underscores the importance of empirical validation and experimentation in determining the most suitable fusion strategy for multimodal learning tasks.

B. Comparison across modalities

We then used the early fusion model to compare the performance of different modalities. Our primary goal was to understand underlying patterns on the modalities' relative strengths and weaknesses. The results are shown in Table II.

TABLE II. COMPARISON OF MODALITIES

Modality	Acc.	F1
Text	0.6796	0.6648
Audio	0.5847	0.5440
Visual	0.4110	0.2752

Fig. 3. Accuracy and F1 scores for each modality on validation and test sets.

Notably, the text modality exhibited the highest accuracy, outperforming both audio and visual modalities. This finding suggests that text data may contain more discriminative or relevant information for the task at hand compared to audio and visual modalities. Possible factors contributing to this discrepancy include the inherent nature of the data, dataset quality, and task-specific requirements. Additionally, it is conceivable that the current models utilised for processing audio and video data may not adequately capture the relevant information present in these modalities. This could stem from factors such as model architectures, feature extraction techniques, or preprocessing methodologies. Furthermore, the complexity inherent in each modality may also play a role, with text data potentially being easier to model due to its direct expression of concepts or categories. Conversely, extracting semantically meaningful information from raw audio or video data may pose greater challenges. These findings underscore the importance of comprehending the unique characteristics and challenges associated with each modality when designing multimodal learning systems. Continued research and innovation are essential to developing effective approaches for processing and integrating diverse modalities in machine learning tasks.

V. MODEL EXPERIMENTATION

We then performed a comparison across models, experimenting with different model architectures to understand their performance on individual modalities. We evaluated four distinct neural network models—TIMNet, Bi-LSTM, CNN, and Transformer—across four modalities: text, audio, visual, and a combined multimodal setting.

A. TIM-Net Model

TIM-Net (Temporal-aware bi-direction Multi-scale Network) [2], introduced in 2023, is a sophisticated architecture designed to enhance Speech Emotion Recognition (SER) capabilities by effectively modelling temporal dynamics in speech data.

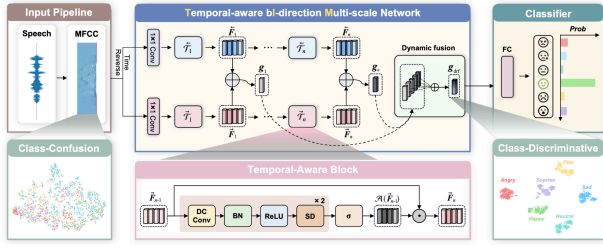


Fig. 4. TIM-Net model architecture [2]

The model processes inputs bidirectionally, integrating past and future contextual information to comprehensively analyse emotional cues. At its core, TIM-Net employs stacked temporal-aware blocks that utilise causal and dilated convolutions. Causal convolutions ensure that outputs are influenced only by previous data points, maintaining the natural order of time, while dilated convolutions expand the receptive field to capture both short-term and long-term dependencies without increasing computational complexity. Outputs from these blocks are dynamically fused and concatenated, providing a rich feature set representing the original sequence processed through various temporal-aware blocks and scales.

B. Bi-LSTM Model

The RNNModel is designed to efficiently process sequential data through its use of bidirectional LSTM layers and a multi-head attention mechanism. It includes a Multi-Head self-attention mechanism, followed by two consecutive Bidirectional LSTM layers, processing the sequence data in both forward and reverse directions.

C. Convolutional Neural Network (CNN)

The CNNModel utilises a series of three 1D convolutional layers with decreasing kernel sizes of 12, 8, and 4, with each layer using ReLU activation to introduce non-linearity. This facilitates a comprehensive extraction of features, from general to specific.

D. Transformer Model

The Transformer Model adapts the transformer architecture, featuring two Transformer blocks, each configured to operate with a model dimension that varies depending on the modality—100 for audio and text, 512 for video, and 712 for multimodal inputs. Each Transformer block contains a self-attention mechanism with four attention heads, facilitating parallel processing of different segments of the data. The use of a large feed-forward network within each block, with a dimensionality of 2048, ensures that the model has sufficient capacity to process complex patterns detected by the attention mechanisms.

E. Model Comparison and Evaluation

The performance metrics employed for the evaluation included accuracy and F1-score, providing insights into both the precision and robustness of each model, as well as an

understanding of their respective strengths and adaptabilities.

TABLE III. MODEL COMPARISON (ACCURACY)

Model	Modality			
	Text	Audio	Visual	All
TIMNet	0.6014	0.3820	0.3395	0.6205
Bi-LSTM	0.7110	0.4972	0.3734	0.6827
CNN	0.6771	0.5471	0.5231	0.6833
Transformer	0.6833	0.5299	0.3395	0.6587

Fig. 5. Accuracy scores for each model and each modality. The best performing model for each modality is bolded.

In terms of accuracy (Table III), the CNN model exhibited superior performance for audio (0.5471) and visual (0.5231) modalities and was notably effective in handling combined multimodal data (0.6833). However, it did not perform as well on the F1 score metric, which is arguably more crucial for a classification task. Conversely, the Bi-LSTM model achieved the highest accuracy in processing text data (0.7110), indicating its strength in capturing long-term dependencies crucial for textual understanding.

TABLE IV. MODEL COMPARISON (F1-SCORE)

Model	Modality			
	Text	Audio	Visual	All
TIMNet	0.4600	0.2337	0.1267	0.5967
Bi-LSTM	0.7009	0.4106	0.1507	0.6813
CNN	0.6630	0.4791	0.4184	0.6712
Transformer	0.6751	0.5019	0.1267	0.6405

Fig. 6. F1 scores for each model and each modality. The best performing model for each modality is bolded.

In terms of F1-scores (Table IV), the TIMNet model demonstrated significantly lower performance across most modalities, particularly in audio (0.2337) and visual (0.1267) contexts. This underperformance could be attributed to the model's original training on datasets characterised by longer timesteps (606), which contrasts sharply with the shorter timestep (110) nature of the current dataset, potentially leading to an overfitting scenario and a lack of adaptability to shorter sequences.

Bi-LSTM again excelled in text modality, achieving the highest F1-score of 0.7009, reinforcing the view that textual data may inherently contain more discriminative features that this model can effectively exploit. For visual data, CNN outperformed other models with a score of 0.4184, affirming its capability in handling spatial features more effectively than its counterparts.

The Transformer model, while not excelling in visual modality, performed best in the audio category with an F1-score of 0.5019. This performance underscores the model's ability to manage complex patterns and relationships within

audio data, possibly due to its attention mechanisms that assess inputs non-sequentially.

Interestingly, the integration of multimodal data did not universally enhance performance across the models, with only the CNN showing notable improvement in this area. This observation may suggest that while combining different types of data can enrich the feature set available to models, not all architectures are equally equipped to leverage this complexity. The CNN's architecture, which inherently processes features at multiple scales, may be better suited to exploit this effectively.

Finally, the findings indicate that modality-specific processing pathways are essential for optimising performance, as different architectures cater to the unique characteristics of each modality. For instance, temporal models like Bi-LSTM are more effective with text, while spatial models like CNNs excel with visual and audio data. This suggests a need for targeted strategies when designing models for specific types of data, possibly incorporating hybrid approaches or ensemble techniques to fully capitalise on the strengths of each architecture.

VI. MULTIMODAL MODEL

We attempt to build a multimodal model that leverages the unique strengths of each modality and facilitates meaningful cross-modal interactions to boost predictive accuracy.

A. Model Architecture

The architecture of the Multimodal Model uses distinct processing pathways for each modality—text, audio, and video—before synthesising the information to make predictions.

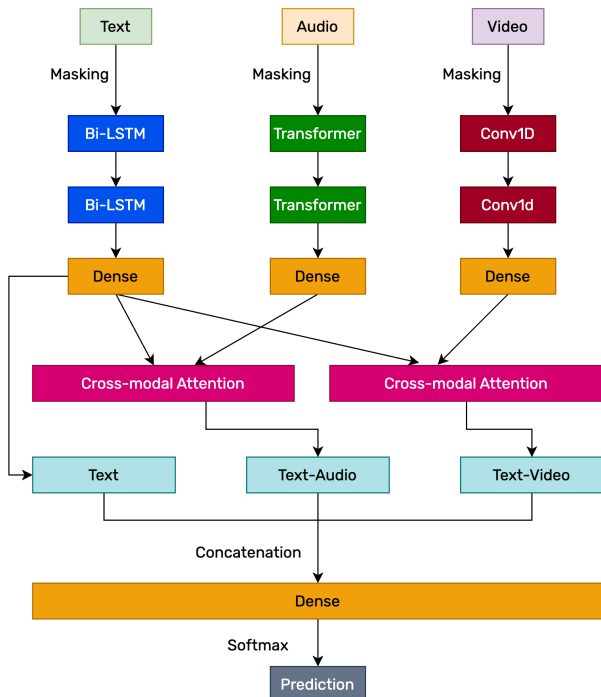


Fig. 7. Architecture for Multimodal model

For text data, the model employs a dual-layer bidirectional Long Short-Term Memory (Bi-LSTM) configuration, which is well-suited for capturing temporal dependencies and nuances inherent in textual information. Each LSTM layer is configured with 64 units and processes data in both forward and reverse directions to maximise the contextual information gleaned from the text. Following LSTM processing, a dense layer with ReLU activation function refines the textual features, preparing them for subsequent integration.

Audio data is processed through a series of Transformer blocks, each consisting of a self-attention mechanism with four heads and a deep feed-forward network with a dimensionality of 2048, followed by a dense layer that aligns the processed audio features with the outputs from other modalities.

Video data processing is handled by a combination of 1D convolutional neural network (CNN) layers and a multi-head attention mechanism. The CNN layers, with kernels sized 8 and 3, respectively, are designed to extract spatial features from the video frames, while the multi-head attention mechanism, configured with eight heads, allows the model to focus on the most relevant features within the video data stream. A subsequent dense layer consolidates these features into a cohesive output.

The model then introduces cross-modal attention layers specifically between the text and the other two modalities—audio and video. These layers are designed to explore and exploit the informative strength of textual data, which has consistently shown to be highly discriminative and robust across various datasets. By focusing on the interactions between text and audio, as well as text and video, the model is able to enhance the relevance of the features extracted from non-textual data.

The final concatenation stage of the model emphasises text features by combining the independently processed text outputs with the outputs from the cross-modal attention layers. This approach not only preserves the integrity of the text data's predictive power but also enriches it with complementary features derived from audio and video inputs. The concatenated features are then passed through a dense layer with softmax activation configured to predict across multiple classes.

This multimodal model architecture is purpose-built to leverage the distinct properties of each data type effectively, ensuring that each modality contributes optimally to the final prediction. By processing each modality independently and then integrating them through sophisticated attention mechanisms and concatenation, the model achieves a balance between modality-specific excellence and cross-modal synergy, setting a robust framework for tackling complex multimodal challenges.

A. Model Performance on IEMOCAP dataset

As depicted in Table 5, the Multimodal Model achieved a high accuracy of 0.7307 and an F1 score of 0.7292, surpassing other comparative models such as TIMNet, Bi-LSTM, CNN, and Transformer. Specifically, the model not only outperformed the TIMNet model, which registered an accuracy of 0.6205 and an F1 score of 0.5967, but also demonstrated improvements over the Bi-LSTM and CNN models, which recorded accuracy scores of 0.6827 and 0.6833, and F1 scores of 0.6813 and 0.6712, respectively. The Transformer model, with an accuracy of 0.6587 and an F1 score of 0.6405, also trailed behind the Multimodal Model.

TABLE V. MULTIMODAL MODEL COMPARISON

Model	Acc.	F1
TIMNet	0.6205	0.5967
Bi-LSTM	0.6827	0.6813
CNN	0.6833	0.6712
Transformer	0.6587	0.6405
Multimodal	0.7307	0.7292

Fig. 8. Comparison of model performance based on multimodal inputs. The multimodal model significantly outperforms other models in terms of both accuracy and F1 scores.

An examination of the confusion matrix revealed specific trends in class misclassification (Fig. 8). Notably, emotions such as 'Happy' and 'Sad' were well-classified, indicating the model's robustness in recognizing these distinct emotional states. However, a considerable amount of class confusion was observed with the 'Neutral' emotion, where various true labels were misclassified as 'Neutral.' Furthermore, for true labels classified as 'Neutral,' the predominant misclassifications were into the 'Anger' category, reflecting the dataset imbalance mentioned earlier.

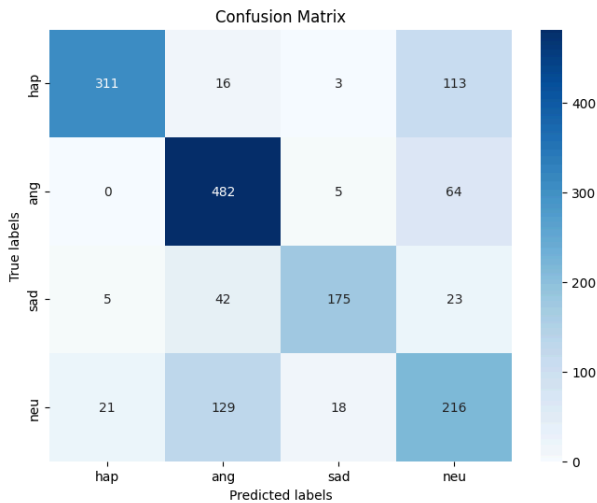


Fig. 9. Confusion matrix for Multimodal model

This pattern suggests a tendency of the model to default to 'Neutral' in ambiguous situations,

which may reflect the inherent challenge in distinguishing subtle emotional cues that differentiate 'Neutral' from more definitively expressed emotions.

To mitigate these effects, future work could explore techniques such as resampling methods, cost-sensitive learning, or the incorporation of synthetic data generation to achieve a more balanced distribution of emotional labels. Addressing the dataset imbalance is paramount to enhancing the model's ability to generalise and perform consistently across all emotional categories.

Overall, the Multimodal Model demonstrates a significant advancement in emotion recognition accuracy and F1 score, as evidenced by its performance on the IEMOCAP dataset. The model's ability to effectively integrate and interpret multimodal data underscores its potential in enhancing the accuracy and depth of emotion recognition systems. The insights gained from the analysis of misclassifications provide valuable directions for further refinement of the model, particularly in improving its sensitivity to less distinctly expressed emotional states such as 'Neutral.'

VII. CONCLUSION

The outcomes of this study emphasise the dependency of model performance on the dataset characteristics, highlighting the inherent complexity of performing comparative assessments of models.

The proposed Multimodal Model, with its modality-specific processing pathways, demonstrates potential in harnessing the strengths of each data type. However, the model's current structure poses limitations in terms of flexibility and adaptability, which are crucial for the creation of a generalizable model in the field of emotion recognition.

Moving forward, this research offers a starting point for further investigation. Firstly, balancing the dataset through methods such as oversampling, undersampling, Synthetic Minority Over-sampling Technique (SMOTE), or the application of class weights is likely to yield more equitable and unbiased models. Additionally, advanced model tuning stands out as an avenue for potential improvement, potentially enhancing the model's capacity to capture and interpret the emotional nuances present in multimodal data. Lastly, validating the model against a broader array of datasets will be instrumental in assessing its robustness and scalability across diverse emotional recognition contexts.

In conclusion, the study underscores the nuanced challenges of emotion recognition and the necessity for continuous refinement of models to achieve a truly empathetic AI, capable of accurately reflecting and responding to human emotions.

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